A parametric leg: from sketch to measured part

M5 • 31 Jan 2027 - 24 Feb 2027 • 75 hours

M05 outcome

During this month, create **a parametric CAD assembly of a leg with three degrees of freedom, drawings, a bill of materials and test specimens**. This is the first physical part of the future walking robot: links, brackets, shafts, fasteners and the cable route. At this stage the leg may be passive, with joints moved by hand; the actuator test rig and force control belong to later modules.

The tasks require you to justify the design: how link length determines part dimensions, which components transfer the load to the body, how to provide access to a screw, how tolerances affect backlash, and why print orientation changes a part's properties. The same approach will later apply to robotic fingers, but designing the complete hand is outside M05.

Learning object and model boundaries

Use three consecutive revolute joints: the first turns the leg in the horizontal plane, and the second and third bend it. In the zero configuration, align the first joint axis with z and the next two axes with y; label the local coordinate frames. Initial link lengths, for example 35/80/100 mm, are only starting geometry. You may change them after checking the layout; these are not the final dimensions of the future robot.

Until a motor is selected, create a separate parametric component representing its envelope. Explicitly mark mounting dimensions that still need confirmation. If the actual actuator has already been selected, use dimensions from its drawing. A provisional motor model does not establish mounting compatibility. A physical passive assembly can use separate shafts and bushings without applying power.

Tools and organisation

Choose one CAD environment: FreeCAD or Fusion if you already know it. Do not learn both simultaneously. Prepare callipers or a ruler, your existing FDM printer, available material, suitable fasteners and a simple fixture for small static loads. Use a validated profile for your printer; the printer model does not automatically determine hole accuracy.

Create cad/native, cad/export, drawings, bom, calculations, coupons and tests. For every print, record the CAD revision, material, orientation, profile, mass and date. M02 covered forces, torques and units; M03 covered ways to describe measurement variability; M04 covered cable-routing requirements. Full kinematics, dynamic walking and actuator operation are not required here.

75 hours and a design strategy

Four blocks from the original programme

A, hours 1-20: 10 h theory + 10 h practice. Sketches, constraints, dimensions and parameters; extrusions, revolutions, patterns, holes and fillets; parts, assemblies, revolute joints and intersections. Deliverable: a link, a bracket and a joint assembly with three labelled axes.

B, hours 21-40: 10+10 h. Fasteners, inserts, bearings, shafts and couplings; fits, datums, tolerance stacks and GD&T fundamentals; cables and maintenance access. Deliverable: drawings, an exploded view, a bill of materials (BOM) and specimens for checking fits.

C, hours 41-60: 10+10 h. Free-body diagrams, equilibrium, load transfer, stress and strain, shear, beam deflection, factor of safety and load uncertainty. Deliverable: a hand calculation for an unfavourable configuration and a comparison of two link versions.

D, hours 61-75: 5+10 h. FEA: restraints, mesh and convergence; FDM anisotropy, orientation and layers; PLA, PETG and TPU; printing and a static test of one assembled leg. Deliverable: a comparison of hand calculations, FEA and measurements, with limitations stated. Total: 35 h of study and analysis and 40 h of practice.

Study rhythm and production

Study the theory in two-hour sessions; split the final 5 h into 2+2+1. Divide each ten-hour practical block into five two-hour sessions. Print small test specimens first, then the critical assembly, and print the entire leg only after checking the fits. Waiting for a print to finish does not count towards study hours; prepare drawings, calculations or the log while the printer runs.

Dates come from the overall plan. Sessions from an adjacent module may occur at the M05 boundary; rest days, holidays and integration stages remain in place. Allocate work hours sequentially across available sessions, without counting hours twice or adding hours. Plan printing in advance so that verification does not depend on the final night of the month.

Priorities when time is limited

First ensure that geometry rebuilds reliably and that connections are justified, then address strength and maintenance access, and finally appearance. If you fall behind, omit photorealistic images, elaborate enclosure styling and a second CAD environment. Retain the three-joint assembly, critical drawings, BOM, hand calculation, a simple FEA check and physical specimens.

Before testing, record the calculated load range and the fixture's permissible loads; a previous result does not automatically carry over after a change in material, layer direction or geometry. No factor of safety in this plan is a universal approval: select it for a specific scenario and support it with data. Conduct a static test on a restrained specimen, without an operating actuator.

Sketches and parameters: build an adjustable link

What to study in CAD

Study the sketch plane, origin, construction geometry and constraints: coincidence, horizontal and vertical alignment, parallelism, perpendicularity, tangency, symmetry and dimensions. Identify the degrees of freedom that remain. A fully constrained sketch should express design intent; fixing every point indiscriminately conceals relationships and makes changes difficult.

Separate driving dimensions from reference dimensions. Use the names L_link, wall, shaft_d, bearing_d, bolt_pitch and cable_clearance. Store units and comments. Practise extrusion (extrude/pad), revolution (revolve), pattern, hole and fillet operations on features of the intended part. Build the main volumes and holes first, then secondary fillets, so the cause of errors is easier to locate when the shape changes.

Task 1. Parametric link

Inputs: the distance between axes, the envelope of the future mount and the selected printing process. Sketch a link with two joint centres, a symmetrical outline and parametric thickness. Extrude the body, add holes and one lightening cavity. Create three length variants by changing only the parameter, rather than stretching faces manually.

Save the model in its native CAD format and a table of variants. Check that centre distances match the specification, the sketch is fully constrained and geometry rebuilds without errors after a length change. Edge case: the minimum length makes holes intersect the cavity. Provide a way to detect this error in the model or checklist, and explicitly state the permitted parameter range.

Task 2. Bracket and a revolved part

Model a U-shaped bracket and a spacer bushing. Use a revolved profile for the bushing and a pattern for repeated fastener holes. Identify mating surfaces, where a flat seat is needed under a screw head, and which tool must reach the fastener. Do not make a wall thin simply because CAD can construct it.

Save both parts in native CAD format, a bracket section view and the results of checking the main dimensions. Change wall thickness and shaft diameter, then verify that symmetry is retained. Deliberate error: a sketch references a face that disappears after a fillet change. Move the reference to a stable datum plane or axis and explain why the revised model tolerates edits better.

Three joints, assembly and maintenance

A component differs from a body

Understand the difference between a part, a component, an instance and an assembly. Two identical bushings in an assembly should reference one model unless one needs a separate revision. Define a fixed base and three revolute joints. Each joint has an axis, a zero position, a positive angular direction and geometric motion limits.

Do not infer mechanical feasibility from a single chosen configuration. Collisions can appear during combined rotations, and the motor housing and screw heads occupy space. Add simple envelopes for the tool, cable and connector as separate reference components; include them in access checks, but exclude them from the printed part's mass.

Task 3. Passive 3-DOF leg

Assemble the base bracket, three joint units and links. Name the joints J1/J2/J3 and label their local axes and initial angles. Check each joint's motion separately, then combinations of extreme positions with intermediate points. Initially, geometry determines the learning model's ranges; the real actuator's limits must be reconciled later.

Save the assembly, a configuration table and screenshots of intersections before and after correction. Check for exactly three intended degrees of freedom, a fixed base and no unintended motion in rigid groups. A discrete set of poses does not prove there are no collisions between them; state the checking increment and investigate suspicious intervals in greater detail.

Task 4. Cable routing and replacement of one joint

Route a provisional cable from the base to the distal joint, allowing extra length for motion, securing it and providing connector access. Check extreme poses: the cable must not become taut, enter a joint gap or rub against a sharp edge. Take the minimum bend radius from the actual cable; for an envelope sketch, mark it as an assumption requiring confirmation.

Describe disassembly in the reverse order of assembly: which screw is removed first, and whether the shaft can be withdrawn and the bushing replaced without dismantling the whole link. Save five maintenance steps and an exploded view. Edge case: the screw head is accessible, but a hex key cannot be inserted along its axis. Add the tool envelope and modify the mounting. For the future hand, this exercise will recur within the smaller space of a finger.

Fits, datums, fasteners and tolerance stacks

Basic machine elements

Study the functions of screws, nuts, washers, threaded inserts in plastic, bearings, bushings, shafts and couplings. Separate their functions: a screw clamps, a shaft defines rotation, and a bearing transfers load while allowing specified motion. The thread of an ordinary screw is generally unsuitable as a precision rotational bearing surface. Select fits to suit the assembly method and actual part tolerances.

A datum is a surface or axis from which important dimensions are specified. GD&T fundamentals explain why axis position and surface perpendicularity differ from a single linear tolerance. Choose one clear set of datums A/B/C, identify the controlled axis and explain how to measure it. Studying the complete geometric tolerancing standard is outside the month's scope.

Task 5. Specimen for checking holes and inserts

Build a small specimen containing a series of fit holes with sizes close to the nominal size of the selected shaft or insert. Choose the increment relative to the expected accuracy of the printer and measuring tool; for example, 0.1 mm may serve as an exercise increment, rather than a promised FDM accuracy. Print the specimen in two orientations with the same material and profile.

Measure the holes using an available method, then check the fit of the actual mating part. For small internal diameters, assess the limitations of calliper measurements and use a known shaft as a functional gauge. Save nominal/measured/fit_result. Edge cases: a hole becomes oval because of layer direction, an insert splits the wall, or a part fits only after undocumented reworking. Record any machining or finishing separately.

Task 6. Axial clearance and drawing

Build a tolerance stack for the assembly: bracket width, two walls, bushings, the rotating link and washers. Calculate minimum and maximum axial clearance for unfavourable tolerance combinations. Do not add standard deviations as though they were guaranteed dimensional limits: the probabilistic estimate in M03 and a worst-case tolerance check answer different questions.

Prepare a drawing of the critical link with views, a section, datums, hole dimensions, material, print orientation and revision. Acceptance check: another person understands what to manufacture and measure. Deliberate error: the dimension chain permits negative clearance and binding after tightening. Correct the spacer bushing or geometry; simply loosening the screw without a controlled fastening arrangement is not a solution.

Statics and strength: how loads are transferred

Essential mechanics

Obtain a free body by mentally separating a part and replacing constraints with reactions. In a static case, the sums of forces and moments are zero. Show how the load passes from the foot through links, shafts and the bracket to the body. Stress is internal force per unit area, and strain is relative dimensional change; modulus E relates them in a simple linear elastic model.

Study tension, compression, shear, bending and deflection. For an ideal cantilever beam with end force F, $M=F\cdot L$, $\sigma_{max}=M\cdot c/I$ and $\delta=F\cdot L^3/(3EI)$. Here c is the distance to the outermost fibre, and I is the second moment of area of the section, not a mass moment of inertia. These equations apply for small deflections, the stated supports and an appropriate material model.

In a simple case, factor of safety n is the ratio of a selected material limit to calculated stress. First state the failure mode and the source of the limiting value. For printed plastic, parameters must correspond to layer direction, temperature and loading conditions; an unknown value cannot be replaced by an arbitrarily generous safety factor.

Task 7. An unfavourable leg pose

Choose an assumed future body mass and a load-distribution scenario, marking them as design assumptions. Do not automatically divide the weight by the total number of legs: some supports may carry no load. For three different poses, calculate the force lever arm about each joint and the base-mount reaction. Include the links' own weight if it is already known from the model or weighing.

Save three free-body diagrams, a force and torque table and the selected design load case. Check force and moment balance, dimensions and sensitivity to a load change. This provides a static reference; foot-strike impact and dynamic accelerations need a separate model. Reflect their uncertainty in the selected scenario and safety factor, rather than concealing it behind a claim of reliability.

Task 8. Two link versions

Simplify the critical part of the link to a beam with a known cross-section. Compare versions A and B: change section height or add a rib while retaining a clear load path. Calculate expected deflection, stress and mass; for a rectangle, $I=b\cdot h^3/12$, so the orientation of its height relative to bending matters.

Prepare a calculation note and two CAD model variants. Verify the prediction: doubling the force doubles deflection in a linear model, while changing height has a nonlinear effect. Edge case: a thin wall beside a hole is not represented by the beam idealisation. Mark this region for local FEA and inspection of the tested specimen. A material's catalogue strength is not yet the allowable strength of the printed assembly.

FEA: formulation, mesh and independent verification

Avoiding unjustified confidence

FEA divides geometry into finite elements and approximately solves the equations of the chosen model. Before running it, record the material, constitutive law, loads and restraints. Fully clamping a large surface is usually stiffer than a real bolted or bushed connection; compare one restraint arrangement with a physically plausible alternative.

For the first exercise, use one simple part and a linear elastic model. Confirm the units of E, loads and dimensions. An isotropic approximation of FDM is a limited model suitable for comparing trends; it does not automatically describe interlayer failure or TPU nonlinearity. Do not reach a final conclusion about a plastic part's failure solely from a coloured equivalent-stress map.

Task 9. A beam, then the actual link

First perform FEA on the same cantilever as in task 8, with the same L, section, E, force and support. Compare deflection at a selected point with the hand calculation. If the discrepancy is large, check units, boundary conditions and geometry before complicating the model. Then apply the approach to the actual link and name the metrics in advance: point deflection, support reaction and stress away from an artificial singularity.

Build three mesh refinement levels while retaining the same problem formulation. For each, record element count, mesh size, solution time and metrics. An exercise target is a change of less than 5% in the selected deflection between the final two meshes; this is a convergence criterion for this experiment, not a universal accuracy or permission to operate.

What to save

Save the original analysis model, material parameters, a list of loaded and restrained surfaces, mesh characteristics and the comparison with the hand calculation. Show the exaggerated deformed shape with its scale labelled; without a label, a viewer may mistake visual magnification for actual deflection.

Edge cases: an unrestrained rigid body, incorrect force units, a load applied to a single vertex and excessive local stress at an ideal sharp edge. A stress maximum that grows without converging needs investigation. Do not simply refine the mesh and declare the latest peak a reliable part stress. For M05, recognising the limitation and checking overall deflection and support reactions is sufficient.

Printing, materials and testing the passive leg

PLA, PETG and TPU according to part function

Compare materials in terms of stiffness, temperature response, creep, impact behaviour, layer adhesion and ease of processing. PLA is useful for initial dimensional prototypes; PETG is often used for tougher parts; TPU suits a flexible foot or pad. These are directions for selection, not guaranteed properties of a particular spool. Consult the manufacturer's data and test your own specimens.

FDM orientation affects interlayer bonding, holes, overhangs and the quality of bearing surfaces. Record layer direction relative to expected tension and bending. Infill density is only one parameter: wall thickness, perimeter count, top and bottom layers, and load transfer may have a stronger influence. Do not compare two materials while simultaneously changing the entire geometry and profile.

Task 10. Specimens and a static check

Prepare two identical beam specimens in different orientations. Measure actual dimensions and mass. Place a specimen in a simple fixture, apply a small known force using a weight or force gauge, and measure deflection at one point. Choose the load range from the preliminary calculation and the fixture's and instrument's capabilities; increase load in steps recorded in advance.

At each level, save load, deflection, hold time and residual deformation after unloading. The fixture must retain the part and weight, and the possible drop area must be clear. Stop testing if the fixture shifts, cracking is heard, a crack appears or deflection becomes unexpectedly large. Destroying the finished leg is not required to pass: a measured range and a description of any first signs of damage or unacceptable deformation are sufficient.

From a test specimen to an assembled leg

Choose orientation and profile from the specimen tests, print the critical links and assemble the passive leg. For the static experiment, use a fixture to lock the joints in one calculated pose; do not rely on the resistance of an unpowered motor. Apply a known, limited foot load and measure total deflection, play before loading and recovery after unloading.

Compare the direction and order of magnitude of deflection with the hand calculation and FEA, explaining the contribution of joints and mountings. Prepare CAD revision B to address one observed defect: a fit, access, a rib or cable clearance. Save photographs, the raw measurement table, a force-deflection plot, observations and the decision on the next revision. A successful static test does not establish walking fatigue life or impact strength.

Work sequence: hours 1-40

Block A: a model that can be changed

Hours 1-2. Choose CAD, create the project structure and a three-axis diagram. **Hours 3-4.** Study geometric and dimensional constraints and fully constrain a link sketch. **Hours 5-6.** Study parameters, expressions and units. **Hours 7-8.** Practise extrusion, revolution, patterns, holes and fillets (extrude/revolve/pattern/hole/fillet) on your own parts. **Hours 9-10.** Study components, instances, assemblies and revolute joints.

Hours 11-20 - practice. 2 h: task 1 and three link lengths. 2 h: bracket and bushing. 2 h: 3-DOF assembly. 2 h: extreme poses and intersections. 2 h: cable, tool and saving revision A. Prepare a short dimensional-specimen print at the end if helpful, so it does not delay the next practical session.

Checkpoint at hour 20: change one link's length and wall thickness. The assembly must rebuild, the axes must remain connected, and the unconfirmed motor envelope must retain its provisional status. If rebuilding fails, correct model dependencies first: adding decorative parts only increases the work needed for revision.

Block B: manufacture and maintenance

Hours 21-22. Study fasteners, inserts and the functions of connections. **Hours 23-24.** Study shafts, bearings, bushings, couplings and fit requirements. **Hours 25-26.** Cover datums and tolerance stacks. **Hours 27-28.** Study how to prepare a clear drawing and the basics of geometric tolerances. **Hours 29-30.** Check cables, tool access and the repair sequence.

Hours 31-40 - practice. 2 h: fit specimens and print preparation. 2 h: measurement and functional trial fitting. 2 h: axial tolerance stack and clearance correction. 2 h: drawings and exploded view. 2 h: BOM, assembly sequence and revision control. If specimen printing takes longer, prepare drawings while it runs; do not change fit criteria after seeing the result.

In the BOM, include part_id, revision, quantity, material or product type, dimensional source and method of obtaining the part. Separate printed parts, standard fasteners and provisional components. Totals by item must match the assembly. For every important item, make clear whether it can be manufactured now or requires selection of the actual actuator.

Work sequence: hours 41-75

Block C: calculate before loading

Hours 41-42. Draw free bodies of the links and their reactions. **Hours 43-44.** Calculate moments for three leg poses. **Hours 45-46.** Study stress, strain, shear and modulus E. **Hours 47-48.** Study a cantilever beam and the influence of cross-section. **Hours 49-50.** Describe uncertainties in load and material properties and justify the design safety factor.

Hours 51-60 - practice. 2 h: leg design load case. 2 h: simplified beam and unit checks. 2 h: versions A and B, comparing mass and deflection. 2 h: prototype test fixture and measurement plan. 2 h: preparation for printing critical parts and the calculation package. State the limits of available fixtures before choosing loads.

At hour 60, explain where the most heavily loaded region lies and why. If the answer rests only on a sense that a region looks thin, return to force lever arms and the load path. If geometry is too complex for a hand calculation, choose a clear idealisation and explicitly identify what it omits.

Block D: compare calculation, simulation and measurement

Hours 61-62. Cover FEA boundary conditions, units and material selection. **Hours 63-64.** Study mesh, convergence and limitations of local peaks. **Hour 65.** Compare PLA, PETG and TPU, layer orientation and the static verification plan.

Hours 66-75 - practice. 2 h: FEA of a simple beam and a hand-calculation comparison. 2 h: actual link and three meshes. 2 h: measured specimens, fits and joint locking. 2 h: static experiment on one assembled leg, table and photographs. 2 h: revision B, final drawings, BOM and presentation. Prepare final parts and specimens before these sessions.

If the available FEA environment requires lengthy setup, retain a minimal analysis of one simple part in an available solver and do not move to a full assembly contact model. Measuring one leg and making a justified comparison with a simple model is more useful than an unfinished complex simulation. The three meshes serve the chosen metric, rather than a target number of analyses.

Final independent check: double the assumed load or rotate a rectangular section by 90°. Predict the change in deflection and the most critical region before running the analysis. Then verify the result by hand and with FEA; increase the physical load only if the new range is justified and stays within the prepared test limits.

Acceptance checks: model, documentation and observations

Six required checks of errors and limitations

1. Rebuilding. Changing length or thickness does not break the main geometry. **2. Collision.** An extreme combined pose has been checked with fasteners and tools included. **3. Fit.** An unfavourable tolerance stack does not conceal binding; retain an unsuitable specimen with the recorded observation.

4. FEA units and supports. Incorrect scale or an unrestrained body is detected through independent verification. **5. Mesh.** A growing local peak is not automatically accepted as a reliable physical result; the selected deflection has been checked for convergence. **6. Physical specimen.** Residual deformation, play or a crack is recorded and influences the next revision, rather than disappearing from the report.

What to prepare

Save the leg assembly in native CAD format, a STEP export and print files with units and revision; critical-part drawings; an exploded view; BOM; parameter table; force, moment and beam calculations; FEA file and settings; mesh table; print profiles; specimens and actual measurements. Photographs must make it possible to identify the physical part by its part_id and layer direction.

Check that all driving sketches are constrained, the assembly has three intended degrees of freedom, the assembly and maintenance sequence is known, and calculation assumptions are not presented as properties of the finished robot. For every deflection or load value, identify its source: hand calculation, simulation or experiment. Compare trends and scale under identical conditions and explain discrepancies.

Self-check questions and handover

How does a geometric constraint differ from a dimension? Why is a part instance preferable to a copied independent body? Where are the datums for the critical hole? What happens to clearance during tightening? Why does support on a threaded surface fail to provide precise rotation? How does the foot load reach the body? Why does section height enter I to the third power?

What does a factor of safety mean when printed strength is unknown? Why does isotropic FEA not establish interlayer strength? How do you distinguish a singularity from a real stress concentration? What did the specimen reveal that CAD did not? Which conclusions cannot be carried over from a static test to walking?

Pass mounting dimensions, envelopes, cable route, mass, maintenance constraints and the test fixture to M06-M07. Once the actual actuator is selected, its mounting must be checked and its loads recalculated. For the future hand, retain the parametric bushing, bracket and fit-checking method; the complete hand and tactile elements will appear in the relevant module and a separate optional block.

Assigned reading: a focused reading route

Read only the specified sections before the corresponding experiment. Most literature is in English; keep explanations, the glossary and the report in English for this edition. Reading and preparation are included in the existing 75 hours; you do not need to work through entire books.

Core reading

1. David Roylance, Mechanics of Materials, open MIT modules. Introduction to Elastic Response; Stresses in Beams; Beam Displacements. For tasks 7-8, derive the relationship between load, stress and deflection. Then read selected parts of Finite Element Analysis for task 9: a mesh approximates a model, so comparison with the beam is compulsory.

2. Alexander Slocum, FUNdaMENTALS of Design. Chapters 4, 8-10: Linkages; Structures; Structural Conn. & Interfaces; Bearings. For tasks 3-5, study only diagrams relevant to your joint, load transfer and part connections. Record which motions the support allows and which reactions it transmits.

3. Yorik van Havre, A FreeCAD Manual. Modeling for product design; Preparing models for 3D printing; Creating FEM analyses. For tasks 1 and 9 and specimen printing, study the parametric model, mesh export and analysis setup. This is an introductory book; check current commands against the help for your version. If using Fusion, read only the general principles of models and meshes.

Three references for the chosen tools

CAD commands. FreeCAD: Sketcher: Constraints, PartDesign: Body, Tools, TechDraw: Views, Dimensions. Alternative: Fusion Parameters and Joints, Limits.

FEA. FreeCAD FEM: Workflow; SimScale Static: materials, boundary conditions, mesh as an alternative reference. In task 9, record the actual restraints and check mesh refinement.

Print material. Prusa Filament Material Guide: PLA, PETG, FLEX. This compares materials; it is not a ready-made profile for your printer. Establish temperature and operating settings from filament manufacturer data and specimen test results.

Reading deliverables. Prepare one load-transfer sketch, one beam calculation, a table of fit dimensions and a justification of print orientation. Connect these results to the decisions in your CAD revision.

Glossary: terms and definitions

Parametric model. Geometry defined by dimensions, relationships and a history of operations; changing a driving parameter rebuilds dependent features.

Sketch constraint. A geometric or dimensional condition that reduces the independent motions of sketch elements, such as coincident points or parallel lines.

Degree of freedom. An independent coordinate needed to specify an admissible mechanism configuration; the number is determined by joints and imposed constraints.

Tolerance. The permitted variation in a specified dimension or geometric parameter, selected according to function and manufacturing method.

Fit. The nature of a connection between mating parts, determined by the combination of their limiting dimensions; a fit may be clearance, interference or transition.

Backlash. Free relative motion when the load direction reverses; it arises from clearances and affects motion transmission accuracy.

Tolerance stack. The relationship between several dimensions that jointly determine a final clearance or position; each deviation's contribution depends on assembly geometry.

Load path. The sequence of components and connections through which a load passes from its application point to the structure's supports.

Stress. Internal force per unit cross-sectional area, measured in pascals. Normal and shear components are distinguished.

Strain. Relative change in material geometry; for uniaxial tension, $\varepsilon=\Delta L/L$, a dimensionless quantity.

Young's modulus. The coefficient relating normal stress and strain in the linear elastic region, measured in pascals.

Stiffness. The ratio of load to the corresponding displacement in a linear model; it depends on material, shape and support conditions.

Strength. A material's or part's ability to sustain load without failure through the mechanism considered; it differs from resistance to elastic deflection.

Second moment of area. A geometric integral describing the distribution of area about an axis; measured in m^4, it influences flexural rigidity $E \cdot I$.

Factor of safety. The ratio of a selected limiting value to an operating value for a specified failure mechanism; it cannot correct an incorrect load model.

Anisotropy. The dependence of material properties on direction; a printed part's strength between layers may differ from its strength along them.

Finite element analysis (FEA). Numerical analysis of a model divided into finite elements; the result depends on material, loads, constraints, mesh and problem formulation.

Mesh convergence. Stabilisation of a selected result as the mesh is progressively refined; it does not establish the correctness of the analysis's physical assumptions.

Equipment, installation and readiness checks

Complete preparation within the first study blocks in place of a repeated general overview. The total remains 75 hours; waiting for delivery is excluded. The statuses below identify actions for you to take, rather than purchases or installations already completed.

USE EXISTING EQUIPMENT

MacBook Pro M4, **your existing 3D printer with a profile for its exact model**, hand tools and a measurement log. There is no need to buy another printer. M05 ends with a passive leg and specimens; the future actuator envelope may still be explicitly provisional.

BUY BEFORE MEASUREMENT AND PRINTING

1 pair of 150 mm callipers with external and internal jaws, a depth gauge and specified measurement error appropriate to the part tolerances. Digital display resolution is not accuracy. Check zero and repeatability on one specimen before inspecting fits.

Material: if supplies are unavailable, 1 spool of PLA for specimens and geometry; PETG if selected; a small amount of TPU only for a flexible part or pad. Filament diameter and settings must suit the printer; calculate the required mass with the slicer, allowing a reserve for repeat specimens.

Fasteners and supports: one set for one passive leg and specimens, according to the released BOM: screws, nuts, washers, threaded inserts, shafts, bushings or bearings. Specify exact diameters and quantities from your CAD model, ensuring that tools can reach the fasteners. Do not order final actuators or a set of parts for the entire robot here.

INSTALL NOW - ONE CAD ENVIRONMENT

macOS ARM: FreeCAD, official macOS arm64 release **or** Fusion if already familiar. FreeCAD requires Sketcher, PartDesign, assembly, TechDraw and FEM; analysis needs CalculiX and Gmsh. Check whether these are included, configure their FEM paths and add missing components following the instructions for your version. FEM Workflow describes the workflow; included components depend on the chosen version's installer.

Install or update only the appropriate slicer and profile for the printer's **exact model**, following its manufacturer's instructions. If choosing Fusion, check access to static FEA beforehand; if unavailable, use a separate simple beam in FreeCAD FEM instead of buying an unnecessary package. Save the CAD, solver and slicer versions and the profile in the project.

INSTALLATION AND READINESS CHECKS

1. Fully constrain a rectangular sketch, extrude it and change a dimension. 2. Export STEP and STL, reopen them and check scale. 3. Create a PDF drawing with a dimension. 4. Analyse a static beam with a known force and compare deflection with the hand formula. 5. Check dimensions and print paths in the slicer, then manufacture one specimen; measure it before printing the whole leg.

Act on the checks: if the specimen does not achieve the required fit or sustain load across the layers, change the print profile or CAD model and produce another specimen. Order subsequent parts from the verified model revision; this result does not automatically call for a different printer or a bulk parts order.

Motion fundamentals: leg constraints and equilibrium

Required, within the 75 hours. Use 1 h from hours 41-42 and 2 h of the hours 51-60 practical work on the leg design load case. Use the existing task 3 assembly and task 7 diagrams. No new equipment or motor experiment is required. Before starting, check yourself: identify the three joint axes and convert mass 0.20 kg into weight 1.962 N at $g=9.81\ m/s^2$.

Freedom of motion requires correctly specified constraints

An open chain of three independent revolute joints with a fixed base has three degrees of freedom. Its configuration is $q=(q_1,q_2,q_3)$. Constraining a part sketch does not immobilise the assembly. If joints are locked by a fixture in a static test, additional constraints and their reactions are introduced. This is a different mechanical problem from a free leg; friction in an unpowered actuator is no substitute for a specified constraint.

Free-body model of one link

For an illustrative special case of task 7, use one horizontal rigid link with $L=0.12$ m, $m=0.20$ kg and its centre of mass at $L/2$. The fixed coordinate frame originates at the left joint; x points right, y up, and positive moments are anticlockwise. Friction and acceleration are absent. The environment applies an upward force $f_{ext}=(0;2)$ N at the right end; weight acts downward. These are exercise values, not a test load for a printed specimen.

Mentally isolate the link. Replace the base with reactions R_x,R_y and the fixture's holding torque τ_{hold}. The environmental force produces moment $+L\cdot 2=+0.240$ N·m; weight produces moment $-(L/2)mg=-0.11772$ N·m. The sum of moments without the fixture is positive, so the free link will not remain horizontal.

Equilibrium requires $\tau_{hold}+0.240-0.11772=0$, giving $\tau_{hold}=-0.12228$ N·m. Force balance gives $R_x=0$ and $R_y+2-1.962=0$: $R_y=-0.038$ N. Here a negative reaction is admissible for a bilateral attachment to the rig; unilateral floor contact cannot realise every calculated force.

Why the Jacobian also appears in statics

For this link at $q=0$, $J=(0,L)^T$. Work done by the external force during a small rotation is $f_{ext}^T\delta p=f_{ext}^T J\delta q$, so the corresponding generalised torque is $J^T f_{ext}=+0.240$ N·m. Holding torque balances it together with the moment of weight. In M05 this is a fixture reaction; in an actuated-joint model, an actuator may fulfil this role. Here f_{ext} is the environment's force on the robot; its sign cannot change while its definition remains the same.

Two deliverables for CAD and the test fixture

1. Mark on the assembly which motions each joint permits and which the fixture prevents. Save two diagrams: the free 3-DOF leg and the restrained test pose. For task 7, identify the physical component that transmits each holding torque.

2. Repeat force and moment balance for the three poses already selected, expressing lever arms and forces in one coordinate frame. Check the moment using both $r\times f$ and $J^T f$ for a simple link. Save a reaction table with units and a load-transfer sketch. A geometrically admissible pose does not establish the structure's ability to sustain load; deflection, strength and dynamics require separate checks.

Assigned reading: Lynch, Park, Modern Robotics: §2.2, Degrees of Freedom of a Robot; §5.2, Statics of Open Chains. Compare the book's sign for force acting on the environment with the f_{ext} convention used here.